
MIRZA EBNUL HASAN

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PROFESSIONAL SKILL

CAD & Design: SolidWorks (CSWA), AutoCAD, Figma

Programming & Data: Python (NumPy, Pandas, Matplotlib), MATLAB, Jupyter Notebooks

Engineering Domains: Aerodynamics & Flow Control, Experimental Setup, Data Acquisition & Analysis

EDUCATION

Texas Tech University - Lubbock, TX

- Texas Tech University — B.S. Mechanical Engineering, Expected May 2028

GPA: 3.750 • Honors College • President's Honor List (Spring 2025) • URS Scholar, FCA Lab

EXPERIENCE

Undergraduate Research Assistant, 08/2025 - Current

Texas Tech University – Lubbock

Flow Control and Aerodynamics (FCA) Laboratory

Supervisor: Dr. Victor Maldonado

- Conducting research on aeronautical flows with a focus on improving aerodynamic, aeroelastic, and aeroacoustic performances.
- Applying passive and active flow control techniques to enhance fluid–structure interaction behavior
- Supporting experimental setup design and data collection for ongoing aerodynamic performance studies

Apprenticeship Trainee, 04/2025 - Current

Red Raider Racing – Lubbock

- Supported shop tasks and documentation for design review.

Student Assistant, 09/2025 - Current

Texas Tech IT Help Central – Lubbock, TX

- Resolved 15–20 technical support tickets per shift (Windows/macOS login issues, imaging, account resets); documented fixes in the knowledge base.
- Reorganized office records (300+ files), reducing retrieval time by ~40%.

IT Intern – Technical Systems & Data Analysis, 06/2025 - 08/2025

Jamuna Bank – Dhaka, Bangladesh

- Automated 10+ data workflows and dashboards in MATLAB/Excel, accelerating report generation by ~35% for the treasury team.
- Drafted 8 standard operating procedures for troubleshooting and secure access provisioning.

Co-Founder, 09/2020 - 01/2025

Next Initiative Foundation – Dhaka

- Coordinated 20+ awareness events (online and in-person) reaching 500+ students, managing logistics and media outreach.
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PROJECTS

- **Hidden Ducted Fan eVTOL Propulsion Integration (CFD + Wind Tunnel Validation) | Texas Tech URS | [AUG 2025 – Present]**

Studied aerodynamic effects of horizontally mounted electric ducted fans (EDFs) placed near an airfoil trailing edge for urban air mobility concepts. Compared EDF placement (top/middle/bottom) and inlet distance to analyze boundary-layer ingestion, drag trends, thrust-efficiency indicators, and wake behavior. Built simulation-to-test workflow

- **Ethical Systems Analysis of Hydrogen Fuel Cell Vehicles (ENGR 2392)**

Built Causal Loop Diagram (10+ variables) modeling adoption feedbacks over 20-year horizon.

Wrote 1,500-word analysis applying prudence (virtue ethics); presented system interventions.

- **HVAC Energy Predictor**

Implemented multi-variable linear regression in Python (NumPy, Pandas, Matplotlib); engineered features from usage logs.

Evaluated model with train/test split & cross-validation; documented results and assumptions.

CERTIFICATIONS

CERTIFIED RISK MANAGEMENT FMEA ISO 31000 EXPERT (SIX SIGMA ACADEMY AMSTERDAM (SSAA))

CSWA – SolidWorks Certified Associate (UDEMY), 07/2025

MATLAB Onramp, 07/2025

